

# The Anchor Protocol:

## A Formally Verified Control Plane for Local Agent Swarms

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March 16, 2026  
Version 1.1 (Port Daddy v3.7.0)

### Abstract

As AI agents transition from isolated copilots to collaborative, autonomous swarms, local development environments face a crisis of coordination and trust. Existing tools designed for human-driven, static infrastructures lack the primitives required to prevent dynamic agents from corrupting shared state or squatting on ephemeral ports.

This paper introduces the **Anchor Protocol**, a cryptographic and semantic identity framework built into the Port Daddy daemon. We detail the protocol’s evolution from symmetric MACs to asymmetric Ed25519 signatures and multi-hop delegation chains (inspired by Macaroons [4]). Furthermore, we present the formal verification of the protocol’s design using the **ProVerif** symbolic analyzer [1] and the mathematical proof of its implementation, demonstrating immunity to algorithm confusion [10], impersonation, and timing side-channels.

**Keywords:** distributed systems security, formal verification, capability-based security, multi-agent coordination, symbolic protocol analysis, ProVerif, JWT security, Ed25519

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## 1 Introduction: The “Local Swarm” Problem

In a multi-agent development environment, agents operate concurrently to read files, spin up test servers, and execute commands. This introduces three critical threat vectors on `localhost`:

1. **Port Squatting (The “Ghost in the Harbor”)**: If Agent A crashes, a malicious or malfunctioning Agent B can bind to Agent A’s previously assigned port, intercepting traffic meant for A.
2. **Resource Contention**: Agents fighting over shared resources (e.g., a SQLite database file) without a centralized locking mechanism cause state corruption.
3. **Privilege Escalation**: A “Reviewer” agent delegated by a “Coder” agent should not possess the rights to initiate production deployments, yet local environments rarely enforce principle-of-least-privilege between processes.

To solve this, Port Daddy introduces the concept of a **Harbor**—a zero-trust, namespace-isolated execution environment where agents must prove their identity and capabilities. Our approach builds upon foundational work in capability-based security by Dennis and Van Horn [7] and the principle of least privilege articulated by Saltzer and Schroeder [15].

## 2 The Anchor Protocol Architecture

The Anchor Protocol defines how agents authenticate to a Harbor and to each other. It issues **Harbor Cards** (highly constrained JSON Web Tokens).

### 2.1 Phase 1: Symmetric Pinning

Early versions of the protocol relied on HS256 (HMAC-SHA256). The primary vulnerability in JWTs is “Algorithm Confusion” (CVE-2015-9235 [11], CVE-2023-48238 [13]), where an attacker modifies the token header to `{"alg": "none"}` or symmetric HS256 while using an asymmetric public key as the HMAC secret.

**Property 2.1** (Algorithmic Pinning). The Port Daddy Verifier employs strict **Algorithmic Pinning**. It entirely ignores the user-provided `alg` header and explicitly forces the underlying crypto library to evaluate the signature using the pre-determined algorithm. This approach is recommended by the JWT Best Current Practices draft [14].

### 2.2 Phase 2: Asymmetric Identity (Ed25519)

To support decentralized Agent-to-Agent (A2A) communication without sharing HMAC secrets, the protocol transitioned to **Ed25519 (EdDSA)** [8, 9]. Ed25519 provides fast, deterministic signatures with small keys and signatures, making it ideal for local agent communication.

**Definition 2.1** (Harbor Key Hierarchy). The Port Daddy Daemon acts as the Root Certificate Authority (CA). Each Harbor is assigned a unique Ed25519 keypair. The Daemon issues Harbor Cards signed by the Harbor’s private key. Agents use the Harbor’s public key (broadcast via Port Daddy’s ambient discovery service) to verify tokens presented by peers.

### 2.3 Phase 3: Stigmergic Delegation (The “Biscuit” Pattern)

Port Daddy v4 introduces multi-hop delegation. When Agent A spawns Agent B, it does not need to request a new token from the Daemon. Instead, it uses **Offline Attenuation**, inspired by Macaroons [4].

**Definition 2.2** (Offline Attenuation). Agent A takes its existing Harbor Card, appends a new set of restricted capabilities (e.g., ["db:read"] reduced from ["db:read", "db:write"]), and signs the *entire chain* with its own ephemeral private key.

The Harbor verifies the Daemon’s root signature, Agent A’s delegation signature, and mathematically ensures that Agent B’s capabilities are a strict subset of Agent A’s.

## 3 Related Work

### 3.1 Formal Verification of Security Protocols

Formal verification of cryptographic protocols has matured significantly over the past two decades. Following the seminal work by Lowe [5] on authentication hierarchies and the Dolev-Yao model [6], several automated tools have been developed:

- **ProVerif** [1]: An automatic symbolic protocol verifier supporting an unbounded number of sessions, used to verify TLS 1.3 [17], Signal [16], and WireGuard.
- **Tamarin** [18]: A state-of-the-art protocol verifier that supports equational properties and inductive proofs, used for analyzing 5G authentication [19].
- **CryptoVerif** [2]: A computationally-sound protocol verifier that provides proofs in the computational model rather than the symbolic model.

Our work builds on these foundations, applying established verification techniques to the novel domain of local multi-agent coordination.

### 3.2 Capability-Based Authorization

Capability-based security dates back to Dennis and Van Horn’s seminal 1966 paper on programming semantics for multiprogrammed computations [7]. Recent work by Birgisson et al. [4] introduced Macaroons, which use chained HMACs for decentralized delegation with contextual caveats. The Anchor Protocol’s delegation chains (Section 5.3) are inspired by Macaroons but adapted for JWT-based identity in local development environments.

### 3.3 JWT Security

JSON Web Tokens have been the subject of extensive security analysis. Algorithm confusion attacks were first systematically studied by McLean [10], with subsequent vulnerabilities documented in CVE-2015-9235 [11] (HS256/RS256 confusion), CVE-2018-1000531 [12] (“none” algorithm), and CVE-2023-48238 [13] (generic algorithm confusion). Our protocol implements the defense recommended by the IETF JWT Best Current Practices [14]: strict algorithm whitelisting and key separation.

## 4 Formal Verification Strategy

Following the industry standard set by AWS (s2n-tls [20]) and Microsoft (Project Everest [22]), we decoupled the verification of the *protocol design* from the verification of the *implementation*.

### 4.1 Symbolic Analysis with ProVerif

To prove the protocol’s logical soundness against a Dolev-Yao adversary [6] (an attacker who controls the entire network), we modeled the Anchor Protocol in **ProVerif** [1]. ProVerif represents protocols as Horn clauses and uses resolution to prove security properties for an unbounded number of sessions.

**Theorem 4.1** (Injective Agreement). For the multi-hop delegation chain, ProVerif confirmed that:

$$\begin{aligned} \text{Accepted}(b, h, \text{cap}) \implies & (\text{IssuedRoot}(a, h, \text{cap}) \wedge \text{Delegated}(a, b, \text{cap})) \\ & \vee \text{IssuedRoot}(b, h, \text{cap}) \end{aligned} \quad (1)$$

This *injective agreement* property [5] guarantees that capabilities cannot be escalated and trust is perfectly transitive.

All three protocol phases were independently verified in ProVerif 2.05. Table 1 summarizes the results; full models are reproduced in Appendices A, B, and C.

| Query                                                                            | Phase           | Model            | Result      |
|----------------------------------------------------------------------------------|-----------------|------------------|-------------|
| $\neg \text{attacker}(\text{master\_key})$                                       | v1 (HS256)      | v1_refined.pv    | <b>TRUE</b> |
| $\text{Accepted} \implies \text{Issued}$                                         | v1 (HS256)      | v1_refined.pv    | <b>TRUE</b> |
| $\neg \text{attacker}(\text{harbor\_sk})$                                        | v2 (Ed25519)    | v2_asymmetric.pv | <b>TRUE</b> |
| $\text{Accepted} \implies \text{Issued}$                                         | v2 (Ed25519)    | v2_asymmetric.pv | <b>TRUE</b> |
| $\text{Accepted}(b) \implies \text{IssuedRoot}(a) \wedge \text{Delegated}(a, b)$ | v3 (Delegation) | v3_delegation.pv | <b>TRUE</b> |

Table 1: ProVerif verification results across all three protocol phases. All queries verified for an unbounded number of sessions under the Dolev-Yao adversary model.

## 4.2 Implementation Verification

A secure protocol is useless if the implementation contains buffer overflows or timing leaks. We extracted the critical JWT parsing and cryptographic comparison logic into a verified core.

**Property 4.1** (Memory Safety). Using the **Kani** Rust Verifier [21], we performed bounded model checking to prove that our parsing logic never panics or accesses out-of-bounds memory, regardless of the input payload.

**Property 4.2** (Side-Channel Resistance). To prevent timing attacks where an adversary guesses signatures byte-by-byte [23], we implemented a custom constant-time byte comparator. Kani formally verified that this comparator is branch-free regarding the contents of the secret arrays.

## 5 Verification Algorithms

In this section, we present the verification algorithms in pseudocode format, following the notation used in protocol verification literature [1, 5]. These algorithms correspond to the formal ProVerif models in Appendix A.

### 5.1 Phase 1: Symmetric HS256 with Algorithm Pinning

Algorithm 1 presents the secure verification procedure that is immune to algorithm confusion attacks (CVE-2015-9235 [11]).

```

1 Require: Pinned Algorithm: hs256
2 Require: Master Key: k_master (secret)
3
4 function Daemon.IssueToken(agent_id, harbor_id, capability):
5     jti <- GenerateNonce()
6     msg <- Encode(agent_id, harbor_id, jti, capability)
7     sigma <- HMAC-SHA256(msg, k_master)
8     emit Issued(agent_id, harbor_id, jti, capability)

```

```

9     return (hs256, msg, sigma)
10
11 function Harbor.VerifyToken(tok, expected_harbor):
12     (alg_header, msg, sigma) <- tok
13     // CRITICAL: Ignore alg_header, pin to HS256
14     if HMAC-SHA256(msg, k_master) != sigma:
15         return _|_ // Invalid signature
16     (a, h, j, cap) <- Decode(msg)
17     if h != expected_harbor:
18         return _|_ // Wrong harbor
19     emit Accepted(a, h, j, cap)
20     return (a, h, j, cap)

```

Listing 1: Algorithm 1: Secure Harbor Verification (HS256 with Algorithm Pinning)

**Security Property:** The algorithm ignores the user-provided  $alg_{header}$  (line 15), preventing algorithm confusion attacks where an attacker sets  $alg = \text{“none”}$  or attempts HS256/RS256 confusion [10].

## 5.2 Phase 2: Asymmetric Ed25519

Algorithm 2 presents the Ed25519-based verification. This phase removes the need for shared HMAC secrets between distributed Harbors.

```

1 Require: Harbor Keypair: (sk_harbor, pk_harbor)
2 Require: Key Distribution Channel: c_key (private)
3
4 function Daemon.IssueToken(agent_id, harbor_id, capability):
5     sk_h <- Receive(c_key) // Get harbor's secret key
6     jti <- GenerateNonce()
7     msg <- Encode(agent_id, harbor_id, jti, capability)
8     sigma <- Ed25519.Sign(msg, sk_h)
9     emit Issued(agent_id, harbor_id, jti, capability)
10    return (msg, sigma)
11
12 function Harbor.VerifyToken(tok, pk_h, expected_harbor):
13     (msg, sigma) <- tok
14     if not Ed25519.Verify(msg, sigma, pk_h):
15         return _|_ // Invalid signature
16     (a, h, j, cap) <- Decode(msg)
17     if h != expected_harbor:
18         return _|_ // Wrong harbor
19     emit Accepted(a, h, j, cap)
20     return (a, h, j, cap)

```

Listing 2: Algorithm 2: Asymmetric Harbor Verification (Ed25519)

**Security Property:** The secrecy of  $sk_{harbor}$  and the unforgeability of Ed25519 [8] ensure that only the legitimate Daemon can issue valid Harbor Cards.

## 5.3 Phase 3: Multi-hop Delegation

Algorithm 3 presents the delegation chain verification, inspired by Macaroons [4].

```

1 Require: Daemon Public Key: pk_daemon
2 Require: Subset Relation: subseq on capabilities
3
4 function Agent.Delegate(token_parent, sk_parent, agent_child, cap_sub):

```

```

5  (msg_parent, sigma_parent) <- token_parent
6  (_, agent_parent, harbor, cap_parent) <- Decode(msg_parent)
7  if cap_sub not_subseteq cap_parent:
8      return _|_ // Cannot escalate capabilities
9  msg_delegate <- Encode_delegation(msg_parent, sigma_parent,
    agent_child, cap_sub)
10 sigma_delegate <- Ed25519.Sign(msg_delegate, sk_parent)
11 emit Delegated(agent_parent, agent_child, cap_sub)
12 return (msg_delegate, sigma_delegate)
13
14 function Harbor.VerifyChain(chain, pk_root):
15     tok_root <- chain[0]
16     (msg_0, sigma_0) <- tok_root
17     if not Ed25519.Verify(msg_0, sigma_0, pk_root):
18         return _|_ // Invalid root signature
19     cap_root <- ExtractCaps(msg_0)
20     pk_current <- pk_root
21     for i <- 1 to |chain| - 1:
22         (msg_i, sigma_i) <- chain[i]
23         if not Ed25519.Verify(msg_i, sigma_i, pk_current):
24             return _|_ // Invalid delegation signature
25         cap_i <- ExtractCaps(msg_i)
26         if cap_i not_subseteq cap_root:
27             return _|_ // Capability escalation detected
28         pk_current <- ExtractPubkey(msg_i)
29     emit Accepted(agent_final, harbor, cap_final)
30     return T

```

Listing 3: Algorithm 3: Delegation Chain Verification

**Security Property:** The subset check on line 7 and line 22 ensures *capability attenuation*—delegated capabilities can only be restricted, never expanded. Combined with Theorem 4.1, this guarantees transitive trust.

## 6 Implementation: Deployed Rust Core

The verified cryptographic core is not a reference implementation—it is the **deployed production code**. The Rust crate `harbor-card-rs` (`core/harbor-card-rs/src/lib.rs`) is compiled to a C dynamic library (`cdylib`) and called from the Node.js daemon via FFI through the `arbiter` module (`lib/arbiter.ts`).

This architecture eliminates the gap between “verified code” and “running code” that weakens many formal verification efforts. The same binary that Kani model-checks is the same binary that the daemon loads at runtime.

### 6.1 Core Verification Logic

```

1  pub fn verify(&self, token: &str, now_ts: i64)
2      -> Result<HarborCardClaims, HarborError> {
3      let parts: Vec<&str> = token.split('.').collect();
4      if parts.len() != 3 {
5          return Err(HarborError::Malformed);
6      }
7      let msg = format!("{}", parts[0], parts[1]);
8      let mut sig_bytes = Self::internal_decode_b64(parts[2])?;
9      let sig_result = self.internal_verify_sig(
10         msg.as_bytes(), &sig_bytes);

```

```

11     sig_bytes.zeroize(); // Scrub signature from memory
12     sig_result?;
13     let mut payload_bytes = Self::internal_decode_b64(parts[1])?;
14     let claims: HarborCardClaims =
15         serde_json::from_slice(&payload_bytes)?;
16     payload_bytes.zeroize(); // Scrub payload from memory
17     if claims.exp < now_ts {
18         return Err(HarborError::Expired);
19     }
20     Ok(claims)
21 }
22
23 pub fn constant_time_compare(a: &[u8], b: &[u8]) -> bool {
24     if a.len() != b.len() { return false; }
25     let mut result = 0u8;
26     for i in 0..a.len() { result |= a[i] ^ b[i]; }
27     result == 0
28 }

```

Listing 4: Deployed Rust Implementation — Harbor Card Verifier (abridged from `core/harbor-card-rs/src/lib.rs`)

Key implementation details: `zeroize` scrubs cryptographic material from memory after use, preventing heap inspection attacks. The constant-time comparator uses XOR accumulation rather than early-return, ensuring execution time is independent of input content.

## 6.2 FFI Bridge to the Node.js Daemon

The Rust crate exports two C-ABI functions consumed by the TypeScript daemon:

```

1  #[no_mangle]
2  pub extern "C" fn harbor_constant_time_compare(
3      a: *const u8, a_len: usize,
4      b: *const u8, b_len: usize
5  ) -> bool { /* ... */ }
6
7  #[no_mangle]
8  pub extern "C" fn harbor_verify_caps_subset_json(
9      root_json: *const c_char, root_len: usize,
10     sub_json: *const c_char, sub_len: usize
11 ) -> bool { /* ... */ }

```

Listing 5: FFI Exports — Called from `lib/arbiter.ts` in the Running Daemon

The daemon’s `arbiter` module binds these at startup:

```

1  constantTimeCompare: lib.func(
2      'bool harbor_constant_time_compare(
3          const uint8_t *a, size_t a_len,
4          const uint8_t *b, size_t b_len)')',
5  verifyCapsSubset: lib.func(
6      'bool harbor_verify_caps_subset_json(
7          const char *root_json, size_t root_len,
8          const char *sub_json, size_t sub_len)')'

```

Listing 6: TypeScript FFI Binding (`lib/arbiter.ts`)

## 6.3 Kani Verification Harnesses

Three bounded model checking proofs are defined in the same source file under `#[cfg(kani)]`:



```

1  #[cfg(kani)]
2  #[kani::proof]
3  #[kani::stub(HarborCardVerifier::internal_pk_from_bytes,
4              stubs::pk_from_bytes_stub)]
5  #[kani::stub(HarborCardVerifier::internal_decode_b64,
6              stubs::decode_b64_stub)]
7  #[kani::stub(HarborCardVerifier::internal_verify_sig,
8              stubs::verify_sig_stub)]
9  #[kani::unwind(10)]
10 fn proof_verify_logic_only() {
11     let pk_bytes: [u8; 32] = kani::any();
12     if let Ok(verifier) = HarborCardVerifier::new(pk_bytes) {
13         let token_bytes: [u8; 32] = kani::any();
14         if let Ok(token_str) =
15             std::str::from_utf8(&token_bytes) {
16             kani::assume(token_str.contains('.')');
17             let _ = verifier.verify(token_str, 0);
18         }
19     }
20 }
21
22 #[cfg(kani)]
23 #[kani::proof]
24 fn proof_constant_time_behavior() {
25     let a: [u8; 16] = kani::any();
26     let b: [u8; 16] = kani::any();
27     let _ = HarborCardVerifier::constant_time_compare(&a, &b);
28 }
29
30 #[cfg(kani)]
31 #[kani::proof]
32 fn proof_capability_attenuation() {
33     let root = vec!["read".to_string(), "write".to_string()];
34     let mut sub = vec!["read".to_string()];
35     assert!(HarborCardVerifier::verify_capability_subset(
36         &root, &sub));
37     sub.push("admin".to_string());
38     assert!(!HarborCardVerifier::verify_capability_subset(
39         &root, &sub));
40 }

```

Listing 7: Kani Proof Harnesses (Deployed Source)

The stubbing strategy deserves explanation: `proof_verify_logic_only` stubs out Ed25519 signature verification and base64 decoding (which involve curve arithmetic that Kani cannot symbolically execute within practical bounds) and exhaustively explores the *parsing and control flow logic*—the split-on-dots, payload extraction, expiry comparison, and error handling paths. This verifies that the logic surrounding the cryptographic primitives never panics, never accesses out-of-bounds memory, and handles all malformed inputs gracefully, regardless of what the cryptographic layer returns.

The build output at `target/kani/aarch64-apple-darwin/debug` contains the CBMC intermediate representations used by Kani’s bounded model checker.

## 7 Security Analysis

### 7.1 Threat Model

We assume a **Dolev-Yao adversary** [6] who:

- Controls the entire network
- Can intercept, modify, and inject messages
- Cannot break cryptographic primitives (cryptography is perfect)
- May corrupt legitimate agents (but not all simultaneously)

This is the standard threat model for symbolic protocol analysis [1].

### 7.2 Security Properties

| Property                             | Verified | Tool     | Deployed  | Reference |
|--------------------------------------|----------|----------|-----------|-----------|
| Master Key Secrecy                   | Yes      | ProVerif | —         | [1]       |
| Algorithm Confusion Immunity         | Yes      | ProVerif | —         | [10]      |
| Authentication (Injective Agreement) | Yes      | ProVerif | —         | [5]       |
| Delegation Integrity                 | Yes      | ProVerif | —         | [4]       |
| Memory Safety (parsing)              | Yes      | Kani     | Yes (FFI) | [21]      |
| Constant-Time Comparison             | Yes      | Kani     | Yes (FFI) | [23]      |
| Capability Attenuation               | Yes      | Kani     | Yes (FFI) | [4]       |

Table 2: Verified Security Properties. “Deployed” indicates the verified Rust code is called by the production daemon via FFI (`lib/arbiter.ts` → `harbor-card-rs`).

### 7.3 Limitations

1. **PID Binding:** The “Ghost in the Harbor” scenario requires binding tokens to the requesting PID, which is handled at the OS level.
2. **Local Transport Security:** On multi-user systems, local unix sockets or loopback TLS should be used to prevent local bearer token sniffing.

## 8 Conclusion

The Anchor Protocol elevates local multi-agent development from a state of “hope-based security” to “math-based security.” By combining:

- Symbolic protocol proofs (ProVerif [1])
- Memory-safe implementation (Rust/Kani [21])
- Capability-based delegation (inspired by Macaroons [4])

Port Daddy provides a formally verified control plane capable of safely orchestrating the next generation of autonomous AI swarms.

## References

- [1] Bruno Blanchet. Modeling and Verifying Security Protocols with the Applied Pi Calculus and ProVerif. *Foundations and Trends in Privacy and Security*, 1(1-2):1–135, 2016. <https://doi.org/10.1561/33000000004>
- [2] Bruno Blanchet. CryptoVerif: A Computationally Sound Security Protocol Verifier. INRIA Research Report, 2007.
- [3] Bruno Blanchet, Vincent Cheval, and Véronique Cortier. ProVerif with Lemmas, Induction, Fast Subsumption, and Much More. In *IEEE Symposium on Security and Privacy (S&P)*, 2022.
- [4] Arnar Birgisson, Joe Gibbs Politz, Úlfar Erlingsson, Ankur Taly, Michael Vrabie, and Mark Lentczner. Macaroons: Cookies with Contextual Caveats for Decentralized Authorization in the Cloud. In *Network and Distributed System Security Symposium (NDSS)*, 2014. <https://doi.org/10.14722/ndss.2014.23212>
- [5] Gavin Lowe. A Hierarchy of Authentication Specifications. In *IEEE Computer Security Foundations Workshop (CSFW)*, 1997.
- [6] Danny Dolev and Andrew Yao. On the Security of Public Key Protocols. *IEEE Transactions on Information Theory*, 29(2):198–208, 1983.
- [7] Jack B. Dennis and Earl C. Van Horn. Programming Semantics for Multiprogrammed Computations. *Communications of the ACM*, 9(3):143–155, 1966.
- [8] Daniel J. Bernstein, Niels Duif, Tanja Lange, Peter Schwabe, and Bo-Yin Yang. High-Speed High-Security Signatures. *Journal of Cryptographic Engineering*, 2(2):77–89, 2012.
- [9] Simon Josefsson and Ilari Liusvaara. Edwards-Curve Digital Signature Algorithm (EdDSA). *RFC 8032*, IETF, January 2017. <https://tools.ietf.org/html/rfc8032>
- [10] Tim McLean. Critical Vulnerabilities in JSON Web Token Libraries. Auth0 Blog, 2015. <https://auth0.com/blog/critical-vulnerabilities-in-json-web-token-libraries/>
- [11] CVE-2015-9235. Algorithm Confusion in jsonwebtoken Node.js library. NIST National Vulnerability Database, 2015.
- [12] CVE-2018-1000531. JWT “none” algorithm vulnerability in jwt library. NIST National Vulnerability Database, 2018.
- [13] CVE-2023-48238. Algorithm confusion in json-web-token library. NIST National Vulnerability Database, 2023.
- [14] Yaron Sheffer, Dick Hardt, and Michael Jones. JSON Web Token Best Current Practices. IETF Draft, 2024. <https://tools.ietf.org/html/draft-ietf-oauth-jwt-bcp>
- [15] Jerome H. Saltzer and Michael D. Schroeder. The Protection of Information in Computer Systems. *Proceedings of the IEEE*, 63(9):1278–1308, 1975.
- [16] Nadim Kobeissi, Karthikeyan Bhargavan, and Bruno Blanchet. Automated Verification for Secure Messaging Protocols and Their Implementations: A Symbolic and Computational Approach. In *IEEE European Symposium on Security and Privacy (EuroS&P)*, 2017.

- [17] Cas Cremers, Marko Horvat, Jonathan Hoyland, Sam Scott, and Thyla van der Merwe. A Comprehensive Symbolic Analysis of TLS 1.3. In *ACM Conference on Computer and Communications Security (CCS)*, 2017.
- [18] Simon Meier, Benedikt Schmidt, Cas Cremers, and David Basin. The TAMARIN Prover for the Symbolic Analysis of Security Protocols. In *Computer Aided Verification (CAV)*, 2013.
- [19] David Basin, Jannik Dreier, Lucca Hirschi, Saša Radomirovic, Ralf Sasse, and Vincent Stettler. A Formal Analysis of 5G Authentication. In *ACM Conference on Computer and Communications Security (CCS)*, 2018.
- [20] AWS s2n-tls. An Implementation of the TLS/SSL Protocols. <https://github.com/aws/s2n-tls>, 2022.
- [21] AWS Automated Reasoning Group. Kani Rust Verifier. <https://github.com/model-checking/kani>, 2022.
- [22] Karthikeyan Bhargavan, Barry Bond, Antoine Delignat-Lavaud, Cédric Fournet, Chris Hawblitzel, et al. Everest: Towards a Verified, Drop-in Replacement of HTTPS. In *Logic in Computer Science (LICS)*, 2017.
- [23] Billy Bob Brumley and Nicola Tuveri. Remote Timing Attacks are Still Practical. In *European Symposium on Research in Computer Security (ESORICS)*, 2011.
- [24] Manuel Barbosa, Gilles Barthe, Karthikeyan Bhargavan, Bruno Blanchet, Cas Cremers, Kevin Liao, and Bryan Parno. SoK: Computer-Aided Cryptography. In *IEEE Symposium on Security and Privacy (S&P)*, 2021.

## A Full ProVerif Models

For completeness, we include the full ProVerif models used for verification. These correspond to the algorithms in Section 5.

### A.1 Phase 1: Symmetric HS256

```

1 (* Port Daddy: Harbor Card v1 (HS256) Formal Model *)
2 type id. type key. type jti. type capability. type alg_type.
3 const hs256_alg: alg_type. const none_alg: alg_type.
4 free c: channel.
5
6 fun hs256(bitstring, key): bitstring.
7   reduc forall m: bitstring, k: key;
8     check_hs256(m, k, hs256(m, k)) = true.
9
10  reduc forall m: bitstring, k: key;
11    verify_generic(m, hs256_alg, hs256(m, k), k) = true;
12    forall m: bitstring, k: key;
13      verify_generic(m, none_alg, m, k) = true.
14
15 fun encode_token(id, id, jti, capability): bitstring [data].
16
17 event Issued(id, id, jti, capability).
18 event Accepted(id, id, jti, capability).
19
20 free master_key: key [private].

```

```

21 query attacker(master_key).
22
23 query a: id, h: id, j: jti, cap: capability;
24   event(Accepted(a, h, j, cap)) ==> event(Issued(a, h, j, cap)).
25
26 let Daemon(k: key, a_id: id, h_id: id, j: jti, cap: capability) =
27   let msg = encode_token(a_id, h_id, j, cap) in
28   let signature = hs256(msg, k) in
29   event Issued(a_id, h_id, j, cap);
30   out(c, (hs256_alg, msg, signature)).
31
32 let SecureHarbor(k: key, expected_h: id) =
33   in(c, (alg_header: alg_type, msg: bitstring, signature: bitstring))
34   ;
35   if check_hs256(msg, k, signature) = true then
36     let encode_token(a: id, h: id, j: jti, cap: capability) = msg in
37     if h = expected_h then
38       event Accepted(a, h, j, cap).
39
40 process
41   new agent_id: id; new harbor_id: id;
42   new token_jti: jti; new secret_cap: capability;
43   (!Daemon(master_key, agent_id, harbor_id, token_jti, secret_cap) |
44    !SecureHarbor(master_key, harbor_id))

```

Listing 8: harbor\_card\_v1\_refined.pv

## B Phase 2: Asymmetric Ed25519

```

1 (* Port Daddy: Harbor Card v2 (Ed25519 / EdDSA) Formal Model *)
2 type id. type skey. type pkey. type jti. type capability.
3
4 free c: channel.
5 free k_dist: channel [private]. (* Trusted key distribution *)
6
7 (** Digital Signatures (Ed25519) **)
8 fun pk(skey): pkey.
9 fun sign(bitstring, skey): bitstring.
10 reduc forall m: bitstring, k: skey;
11   check_sign(m, pk(k), sign(m, k)) = true.
12
13 fun encode_token(id, id, jti, capability): bitstring [data].
14
15 (** Events **)
16 event Issued(id, id, jti, capability).
17 event Accepted(id, id, jti, capability).
18
19 (** Queries **)
20 free secret_key: skey [private].
21 query attacker(secret_key).
22
23 query a: id, h: id, j: jti, cap: capability;
24   event(Accepted(a, h, j, cap)) ==> event(Issued(a, h, j, cap)).
25
26 (** Processes **)
27 let Daemon(a_id: id, h_id: id, j: jti, cap: capability) =
28   in(k_dist, sk_h: skey);

```

```

29   let msg = encode_token(a_id, h_id, j, cap) in
30   let signature = sign(msg, sk_h) in
31   event Issued(a_id, h_id, j, cap);
32   out(c, (msg, signature)).
33
34   let Harbor(pk_h: pkey, expected_h: id) =
35     in(c, (msg: bitstring, signature: bitstring));
36     if check_sign(msg, pk_h, signature) = true then
37       let encode_token(a: id, h: id, j_1: jti, cap_1: capability)
38         = msg in
39       if h = expected_h then
40         event Accepted(a, h, j_1, cap_1).
41
42   (** Main Process **)
43   process
44     new agent_id: id; new harbor_id: id;
45     new token_jti: jti; new secret_cap: capability;
46     let harbor_pk = pk(secret_key) in
47     out(c, harbor_pk);
48     (
49       (!Daemon(agent_id, harbor_id, token_jti, secret_cap)) |
50       (!Harbor(harbor_pk, harbor_id)) |
51       (!out(k_dist, secret_key))
52     )

```

Listing 9: harbor\_card\_v2\_asymmetric.pv — Verified in ProVerif 2.05

## C Phase 3: Multi-hop Delegation

```

1  (* Port Daddy: Harbor Card v3 (Delegation Chains) Formal Model *)
2  type id. type skey. type pkey. type capability.
3
4  free c: channel.
5
6  (** Cryptographic Functions **)
7  fun pk(skey): pkey.
8  fun sign(bitstring, skey): bitstring.
9  reduc forall m: bitstring, k: skey;
10    check_sign(m, pk(k), sign(m, k)) = true.
11
12  reduc forall c1: capability; is_subset(c1, c1) = true.
13
14  fun base_token(id, id, id, capability): bitstring [data].
15  fun delegate_token(bitstring, id, capability): bitstring [data].
16
17  (** Events **)
18  event IssuedRoot(id, id, capability).
19  event Delegated(id, id, capability).
20  event Accepted(id, id, capability).
21
22  (** Queries **)
23  free harbor_id: id.
24  query b: id, cap: capability, a: id;
25    event(Accepted(b, harbor_id, cap)) ==>
26      (event(IssuedRoot(a, harbor_id, cap))
27       && event(Delegated(a, b, cap)))
28    || event(IssuedRoot(b, harbor_id, cap)).

```

```

29
30 (** Processes **)
31 free daemon_id: id.
32 free daemon_sk: skey [private].
33
34 let Daemon(a: id, cap: capability) =
35   let msg = base_token(daemon_id, a, harbor_id, cap) in
36   let s = sign(msg, daemon_sk) in
37   event IssuedRoot(a, harbor_id, cap);
38   out(c, (msg, s)).
39
40 let AgentA(a: id, a_sk: skey, b: id, sub_cap: capability) =
41   in(c, (msg: bitstring, s: bitstring));
42   let base_token(=daemon_id, =a, =harbor_id,
43     root_cap: capability) = msg in
44   if check_sign(msg, pk(daemon_sk), s) = true then
45     if is_subset(sub_cap, root_cap) = true then
46       let d_msg = delegate_token((msg, s), b, sub_cap) in
47       let d_s = sign(d_msg, a_sk) in
48       event Delegated(a, b, sub_cap);
49       out(c, (d_msg, d_s)).
50
51 let Harbor(a_pk: pkey) =
52   in(c, (d_msg: bitstring, d_s: bitstring));
53   let delegate_token((base_msg: bitstring, base_s: bitstring),
54     b: id, sub_cap: capability) = d_msg in
55   let base_token(=daemon_id, a: id, =harbor_id,
56     root_cap: capability) = base_msg in
57   if check_sign(base_msg, pk(daemon_sk), base_s) = true then
58   if check_sign(d_msg, a_pk, d_s) = true then
59   if is_subset(sub_cap, root_cap) = true then
60     event Accepted(b, harbor_id, sub_cap).
61
62 (** Main Process **)
63 process
64   new sk_a: skey;
65   let pk_a = pk(sk_a) in
66   out(c, pk_a);
67   new id_a: id; new id_b: id; new cap_root: capability;
68   (
69     (!Daemon(id_a, cap_root)) |
70     (!AgentA(id_a, sk_a, id_b, cap_root)) |
71     (!Harbor(pk_a))
72   )

```

Listing 10: harbor\_card\_v3\_delegation.pv — Verified in ProVerif 2.05